Understanding the learning of disabled students: An exploration of machine learning approaches

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***Abstract*—** *Research is limited in demonstrating how learning analytics can support students with a learning disability. It is vital to scrutinise the demographics of students with and without disabilities and their effect on performance. Machine learning algorithms can provide valuable insights by mining this expanding education data. This study aims to analyse the relationship between disabled and non-disabled students' demographic with their scores and the number of attempts to complete a module to develop prediction models. Three models - Adaptive Boosting, Random Forest, and K nearest neighbour - were used. The results found that the Adaptive boosting algorithm delivered the highest prediction accuracy.*

Keywords*—Learning Analytics, Learning Disability, Demographic, Prediction, Adaboost, Random Forest, kNN, Machine Learning*

I. INTRODUCTION

As technology is developing at a tremendous pace, the use of technology in learning has also developed over time. Along with this, Educational Data Mining (EDM) advancements have seen the rise of learning analytics (LA) as a related field [20]. These developments make it plausible to identify at-risk students, predict their scores, build adaptive learning environments, and facilitate personalised feedback. It enables to development of the quality of education [22].

Despite these developments, little consideration has been given to students with learning disabilities [11]. The term "learning disability" is interpreted differently in different countries. In Australia, the term learning disability refers to students with severe difficulty in literacy and numeracy; most of the time, learning disabilities are regarded to have a neurological basis [14]. Dyslexia, dysgraphia, and dyscalculia are a few of the learning disabilities. In most cases, students do not feel it safe or necessary to disclose their disability [13]. At the same time, learning-disabled students suffer from issues that can impede their learning, such as stress and anxiety, lack of self-esteem, poor study-life balance, personal relationship problems, and financial pressures [16]. Therefore, there is an urgent need to identify students' learning disabilities in a way that allows them to feel safe and support their learning in response to specific disabilities.

For this purpose, developing a fair machine learning model is crucial; [19], however, such models are sparse in the literature. Moreover, the accuracy of predictive models is often in question [19]. In this paper, we take a first step toward developing a fair model for predicting students' on a publicly available dataset, Open University Learning Analytics Dataset (OULAD), Open University, as we could not find any dataset available in the Australian context. This paper uses three prediction models to identify relationships between the demographic factor (disabled and non-disabled students) scores on different assessments. The results from our exploratory analysis provide an initial insight into the learning of students with learning disabilities.

II. LITERATURE REVIEW

In this section, we will discuss the difficulties faced by students with learning disabilities and the machine learning approaches that have been taken to acknowledge and resolve these issues.

A. Learning Disability

Conventional teaching practices in physical and online learning environments fail to consider students with learning disabilities, which creates significant issues for these students. As a result, students with dyscalculia – difficulty with counting, dyslexia – and students with difficulty in reading some specific fonts are left behind [21]. The rapid digitisation of learning technology has failed to consider the compatibility of current education practices with disabled students compared to students who do not have such challenges. For example, while uploading lectures online, not having a text description or accessibility in reading-specific fonts can add more challenges to learning for learning-disabled students [11]. As a result of not receiving enough support, children face difficulties throughout adulthood and perform less than optimally at universities. Additional consequences can occur when students take anti-depressant or anti-anxiety medication [1]. Learning analytics offers a promising way to support these students by drawing on individual learning data and providing timely interventions. However, for this to happen, practical machine learning approaches are needed.

B. Machine Learning Approaches

With advances in EDM and LA, various machine learning approaches have been widely employed [3]. Three approaches have been discussed in the literature. The random forest algorithm has proved to be more than 81% accurate in

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predicting students' performance using the OULAD dataset [9]. The random forest can be used in batch and adaptive modes for predicting students' success with the expanding education data. AdaBoost has delivered promising results of 79% at the end of the first semester and increased its accuracy to 90% at the end of the semester [10]. Using multilayer perception, AdaBoost has 87.6% accuracy in classifying and predicting the performance of students enrolled in software engineering courses [4]. Anxiety prediction, with an accuracy of 0.943, among the students was performed using the AdaBoost machine learning model [6]. Although KNN has not provided promising accuracy in predicting student performance, it can be used as a classification algorithm, and the accuracy varies depending on the dataset [7].

III. Research Questions

This study explores the possibilities of a machine learning approach to support students with disabilities. Three research questions guided this study:

- RQ 1: Is there any relationship between disabled and non-disabled students' demographics and their scores in different assessment types?
- RQ 2: Is there any relationship between disabled and non-disabled students' demographics with the number of attempts they take to complete a module?
- RQ 3: Is there a difference in the accuracy of predictions made by different machine learning algorithms?

IV. Methodology

In this paper, we focused on three algorithms, Adaboost, Random Forest, and kNN, and compared their prediction accuracy with MSE, RMSE, MAE, and R-squared values. The model-building process was performed in the Orange Data Mining application. The models were developed in five steps. Starting from step 1 Data Collection, exploratory data analysis & Data preparation, selection & training of the model, calculating model accuracy, and lastly analysing and delivering the results fig[1].

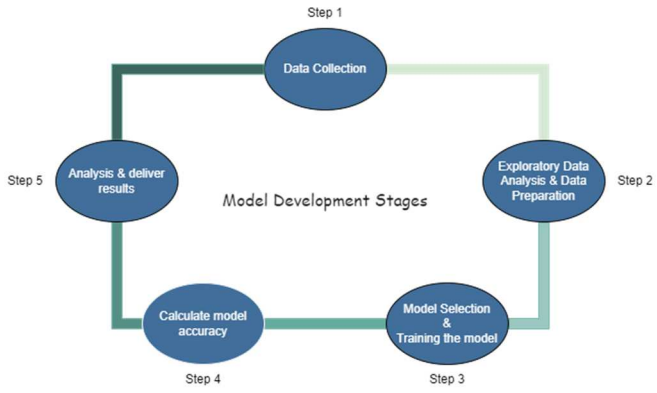


Fig 1: Model Development Stages

A. Dataset

The dataset for this study was an Open University Learning Analytics Dataset (OULAD) developed by Open University from their virtual learning courses in 2013 and 2014. The most valuable fact of this dataset is that it contains the demographics of 32,593 students and their outcomes in 22 courses, including the logs of their interactions [15]. For this study, only the following variables were considered from the dataset: demographics (age, gender, highest education, disability); a number of previous attempts; final results and assessments; assessment score; assessment type (Computer Marked Assessment (CMA), exam, and Tutor Marked Assessment (TMA)). The demographics were extracted from the studentInfo table, which contained 32,593 rows. The studentAssessment table contained 173,912 records [15].

B. Data Preparation

Data pre-processing was conducted in Python, while visualisations were performed in Tableau. Outliers and any presence of N/A values were checked in all three tables. In order to answer our research questions, we divided the table into 1) details of students with / had declared disability; and 2) details of students without / had not declared any disability. To investigate relationships between the disability factor and scores in different assessment types and on a number of previous attempts, the gender and highest education of the disabled and non-disabled students were analysed at the next stage using exploratory data analysis.

C. Visualisations

1) RQ1: Relationship between disabled and non-disabled students' demographics and their scores in different assessment types: Gender vs Scores along with assessment type: Figure [2], [3] illustrates a higher score for female disabled students than male disabled students in Teacher Marked Assessment (TMA) assessment, 26.95 %. On the other hand, male non-disabled students scored more in TMA 32.11%. From fig [2] and [3], female disabled students scored higher in the assessments than male disabled students. Contrastingly male non-disabled students were visualised to perform better than the female students. This visualisation supports the relationship between disability and gender scores in different assessments.

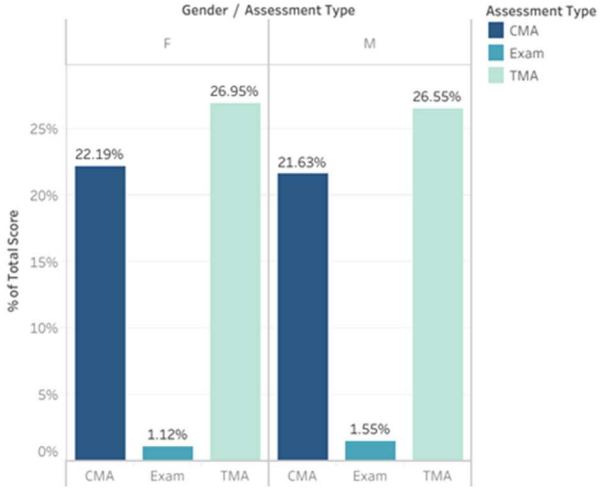


Fig 2: Gender vs Percentage of total score in each assessment type – Disabled students

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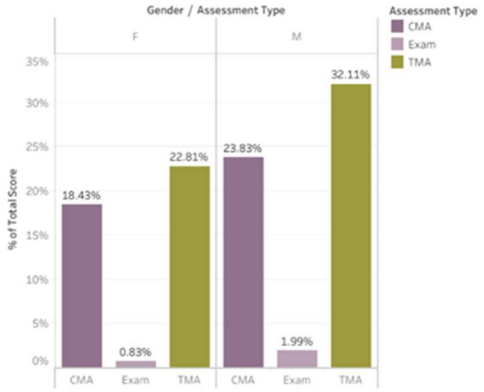


Fig 3: Gender vs Percentage of total score in each assessment type - Non-Disabled Students

a) Gender vs Number of previous attempts: As shown in Figure [4], among the disabled students, males took more attempts (19.83%) than female disabled students to pass a module (18.79%). Similarly, for non-disabled students, males attempted most (27.60%), to pass modules, fig[5]. Withdrawal rate with attempts was greater for disabled students - female 12.10% , male 13.45%, than for non-disabled - female 6.49% , male 9%, fig [4], [5].



Fig 4: Gender vs Number of previous attempts along with final result - Disabled Students

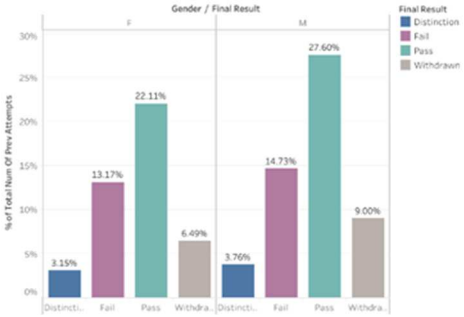


Fig 5: Gender vs Number of previous attempts along with final results - non-disabled students

2) RQ 2. Relationship between highest education and number of previous attempts: From both fig [6] , [7], the number of attempts taken by disabled students was highest, 49.63% for "Lower than A level" students. For non-disabled students, "A level or equivalent" students had taken more attempts, 44.23%, than other students.

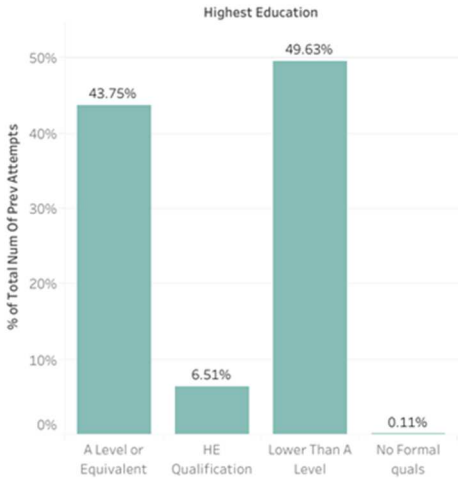


Fig 6: Highest Education vs % of the total number of previous attempts- Disabled Students

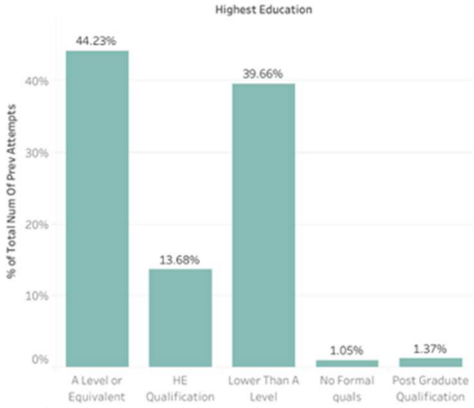


Fig 7: Highest education vs % of the total number of previous attempts - non-disabled students

D. Adaptive Boost Algorithm (AdaBoost)

Adaptive Boosting or AdaBoost, one of the most effective ensemble learning algorithms, works by aggregating all the weak classifiers, generating a strong one [8] and sustaining an adaptive weight distribution [18]. A weighted accumulation that represents the outcomes of the weighted classifier is linked with the outputs of the weak students. It works both for classification and regression. The algorithm,

$$F_T(x)= \sum_{t=1}^{T} f_t(x) \qquad (1)$$

every f_t is a classifier, weak student, that outputs the predicted class in relation to input x. Weak students produce prediction outputs, t = for every iteration Et = sum of training error.

$$E_t= \sum_{i} E\,[F_{t-1}(x_i)+\alpha_t h(x_i)] \qquad (2)$$

In this case, E(F) is the error function. AdaBoost tweaks the weak students and improves the individual results.

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E. Random Forest Algorithm

Random forest, another ensemble classification technique to enhance classification accuracy, generates a number of decision trees that are trained on various portions of the same training set [5]. To create the K number, a random forest randomly chooses the attributes. The error rate determines the strength of the tree; the stronger the tree, the stronger the forest will be. Random Forest can handle categorical variables and missing values, works effectively on large datasets, do not require pruning, and never overfits [2]. Random forest takes 2/3 of the original data, and the rest are used for classifying unbiased errors, called out-of-bag (oob). Gini Index is used to determine the final class in each tree. Each class's Gini index is determined by subtracting the sum of squared probabilities. As defined in [5], the Gini index equation is,

$$\text{Gini}(t)=1-\sum_{i=0}^{c-1}[p(i|t)]^2 \qquad (3)$$

F. K Nearest Neighbour Algorithm (kNN)

Out of all the machine learning algorithms, kNN is the simplest, having several uses in various industries, including pattern recognition, online marketing, and cluster analysis of image datasets. The k-value specifies how many neighbours are closest. The neighbours are chosen from a group of objects or objects with similar characteristics or values [17]. One advantage of kNN is that it is easy to understand and has a fast-training rate [7]. Euclidean's distance is calculated to calculate the length as given [17],

$$d(A, B)=\sqrt{(x_1-x_2)^2+(y_2-y_1)^2} \qquad (4)$$

G. Performance Measures

The essential part of developing a machine learning model is determining its accuracy. Mean Squared Error (MSE), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R squared were used to determine the accuracy of the model's prediction. MSE illustrates the average squared difference between the dataset's original value and the model's predicted value. RMSE is simply the square root of MSE. MAE, on the other hand, measures the average of the residuals. R-squared represents the proportion of the variance. The lesser the MSE, RMSE, and MAE value, the more accurate the model. A higher R-squared value represents a better model for the particular dataset, which usually falls between 0 – 1. The mathematical background is calculated as defined in [12]:

$$\text{MSE}=\frac{1}{m}\sum_{i=1}^{m}(X_i-Y_i)^2 \qquad (5)$$

$$\text{RMSE}=\sqrt{\frac{1}{m}\sum_{i=1}^{m}(X_i-Y_i)^2} \qquad (6)$$

$$\text{MAE}=\frac{1}{m}\sum_{i=1}^{m}|X_i-Y_i| \qquad (7)$$

$$R^2=1-\frac{\sum_{i=1}^{m}(X_i-Y_i)^2}{\sum_{i=1}^{m}(Y-Y_i)^2} \qquad (8)$$

V. Analysis

A. RQ 3: Prediction Accuracy of the algorithms

1) Disabled Students: AdaBoost produced the highest accuracy for both gender and demographic vs score dataset. From Table 1, for gender vs score, the R squared value was the highest in the AdaBoost algorithm, 0.998. The MSE, RMSE, and MAE values were also the lowest, 0.739 MSE, 0.204 for RMSE, and 0.204 for MAE for this algorithm. For the higher education vs score dataset, the AdaBoost model provided the highest R squared value, 0.998 and the lowest MSE: 0.749, RMSE: 0.865, and MAE: 0.211. For the other two datasets, gender vs a number of previous attempts and highest education vs a number of previous attempts, random forest showed the highest accuracy. In gender vs number of previous attempts' accuracy, the random forest has the highest R squared value among the other models, 0.787. This model also has the least MSE: 0.067, RMSE: 0.260, and MAE: 0.111. In highest education vs number of attempts, the random forest provided the leading R squared value, 0.789. This model has also lowest MSE: 0.067, RMSE: 0.259, and MAE: 0.111 values. Both AdaBoost and Random Forest models have illustrated satisfactory accuracy. kNN on the other hand demonstrated the lowest accuracy.

Table 1: Prediction Accuracy Measurement – Disabled student

Disabled Students					
	Model	***MSE***	***RMSE***	***MAE***	***R squared***
Gender Vs Score	KNN	151.192	12.296	8.670	0.639
	Random Forest	74.564	8.635	5.776	0.822
	AdaBoost	0.739	0.204	0.204	0.998
Gender Vs Number of attempts					
	AdaBoost	0.071	0.266	0.113	0.777
	Random Forest	0.067	0.260	0.111	0.787
	kNN	0.074	0.272	0.113	0.766

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Highest Education Vs Score	AdaBoost	0.749	0.865	0.211	0.998
	Random Forest	74.154	8.611	5.806	0.823
	kNN	151.211	12.297	8.670	0.639
Highest Education Vs Number of attempts					
	AdaBoost	0.068	0.261	0.110	0.784
	Random Forest	0.067	0.259	0.111	0.789
	kNN	0.074	0.272	0.113	0.766

2) Non-disabled Students: As shown in Table 2, AdaBoost reported the highest accuracy among all the other models in every scenario. For example, in gender vs score, AdaBoost had the R squared value, 0.998, and the lowest MSE: 0.868, RMSE: 0.932, and MAE: 0.236. For gender vs the number of previous attempts, AdaBoost produced a 0.820 R squared value. The highest education vs score and highest education vs the number of previous attempts were reported consecutively, 0.998 and 0.816, which were the leading R squared values. Random Forest for this non-disabled students' dataset also performed satisfactorily. On the contrary, kNN reported the lowest accuracy among all the models.

Table 2: Prediction Accuracy Measurement - Nondisabled Students

Non Disabled Students					
	Model	***MSE***	***RMSE***	***MAE***	***R squared***
Gender Vs Score	KNN	135.070	11.622	8.116	0.629
	Random Forest	76.475	8.745	6.020	0.790
	AdaBoost	0.868	0.932	0.236	0.998
Gender Vs Number of attempts					
	AdaBoost	0.036	0.190	0.059	0.820
	Random Forest	0.039	0.197	0.065	0.808
	kNN	0.041	0.203	0.063	0.796
Higher Education Vs Score					
	AdaBoost	0.794	0.891	0.223	0.998
	Random Forest	72.727	8.528	5.816	0.800
	kNN	134.996	11.619	8.114	0.630
Higher Education Vs Number of attempts					
	AdaBoost	0.037	0.192	0.060	0.816
	Random Forest	0.038	0.196	0.065	0.809
	kNN	0.041	0.203	0.063	0.796

VI. Discussion

The analysis of the OULAD dataset found no significant difference between disabled students' and non-disabled students' performance. However, gender and highest education demographics revealed a significant difference between these two demographic students.

It is necessary to scrutinise every aspect to have a clear visualisation of the factors that may affect the performance of disabled students. The number of previous attempts taken was higher for disabled students, and the highest level of education in disabled students was only "lower than A level" for most students. On the other hand, the highest level of education for non-disabled students was "A level or equivalent", fig [7]. This demonstrates the need to emphasise more support for inclusive education. Although the percentage of total scores in different assessment types was not much difference between disabled and non-disabled students, a significant difference in scores were observed when gender was considered. This study also found that different machine learning approaches yielded different results for prediction. In this case, Adaboost showed the highest accuracy for both datasets, followed by random forest. kNN reported the lowest accuracy for our dataset. Nevertheless, as the model's accuracy depends on datasets and actions, kNN can still be used in other prediction models.

VII. Conclusion & Future Work

More thorough research is needed to analyse "learning disability" in higher education. As this was an exploratory study, we acknowledge the possible limitations. Firstly, this study employed a smaller number of demographic and score data. Secondly, the dataset did not elaborate on the type of disabilities students had. Thirdly, there were limitations of the prediction models. Lastly, we did not have publicly available dataset of Australian learners with learning disability. Future work will firstly include applying adaptive learning models with more data to acquire better results and secondly, more emphasize will be given on analysisng the Australian context with accurate data for learninig disability.

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REFERENCES

[1] Brennan, C., Müller, N., Ball, M. J., & Damico, J. S. (2021, March). *Learning disabilities*. ResearchGate. Retrieved July 29, 2022, from https://www.researchgate.net/publication/350029419_Learning_Disabilities

[2] Jaiswal, J. K., & Samikannu, R. (2017, October 19). *Application of random forest algorithm on feature subset selection and classification and regression*. IEEE Xplore. Retrieved July 29, 2022, from https://ieeexplore.ieee.org/abstract/document/8074494?casa_token=18hpIVoIYWQAAAAA%3AGe22CYETks41wLex4G4KqtybqXn5rJOTwLRtwOyUFQELHIxU5uQmnsOPnSM2SCEPBsfi4JzyP3E

[3] Romero, C., & Ventura, S. (2020, January 13). *Educational data mining and learning analytics: An updated survey*. Wiley Interdisciplinary Reviews. Retrieved July 29, 2022, from https://wires.onlinelibrary.wiley.com/doi/abs/10.1002/widm.1355

[4] Abidin, A. F. Z., Darmawan, M. F., Osman, M. Z., Anwar, S., Kasim, S., Yunianta, A., & Sutikno, T. (2019, October 10). Adaboost-multilayer perceptron to predict the student's performance in software engineering. Bulletin of Electrical Engineering and Informatics. Retrieved July 29, 2022, from https://www.beei.org/index.php/EEI/article/view/1432/1245

[5] Ahmed, N. S., & Sadiq, M. H. (2018, November 29). Clarify of the random forest algorithm in an educational field. IEEE Xplore. Retrieved July 29, 2022, from https://ieeexplore.ieee.org/abstract/document/8548804?casa_token=o1JCGpTThuUAAAAA%3ArAI3q3tUakGyHKgOx-Q75IqW83ivDJ9Ln_k8wayW0rl_AwoGyzznHajCXcd13TrkGU5cBi-cpug

[6] Alharthi, H. (2020, October). Predicting the level of generalized anxiety disorder of the coronavirus pandemic among college age students using Artificial Intelligence Technology. IEEE Xplore. Retrieved July 29, 2022, from https://ieeexplore.ieee.org/abstract/document/9277785?casa_token=7e6d_HNS7iYAAAAA%3AT7jewXZz3P3nr0Cc0oLR9bFGy3E9Qe9uTCvENCZh9T6DKKMN5tV67T1oM5NgOeamcNzinSm0ikM

[7] Amra, I. A. A., & Maghari, A. Y. A. (2017, October 23). Students performance prediction using KNN and Naïve Bayesian. IEEE Xplore. Retrieved July 29, 2022, from https://ieeexplore.ieee.org/abstract/document/8079967?casa_token=9DxqsC1mz0kAAAAA%3AmDCAjkvXR- Kraayenoord, C. E. van, & Chapman, J. (2016, March). Learning disabilities in Australia and New Zealand . ResearchGate. Retrieved July 29, 2022, from https://www.researchgate.net/publication/303993211_Learning_Disabilities_in_Australia_and_New_Zealand AWN0ORWqXg9ABdal0x3SiZ_MfFYZDTZW_QFDuvBiXLlo_cgH3ogXu3_Gcu82oly6U

[8] Baig, M. M., Awais, M. M., & El-Alfy, E.-S. M. (2017, March 10). AdaBoost-based artificial neural network learning. Neurocomputing. Retrieved July 29, 2022, from https://www.sciencedirect.com/science/article/pii/S0925231217304241?casa_token=SKjTOY_2rBcAAAAA%3Ai2OYoJHV_XOtWQHqeUuS94c2et69FORobclJdrRdAZYmGZOe0J4m0dHoP1mzv7Kv59aNKqU6jlGr

[9] Batool, S., Nisar, M. W., Rashid, J., & Kim, J. (2021, September). A Random Forest Students' Performance Prediction (RFSPP) Model Based on Students' Demographic Features. ResearchGate. Retrieved July 29, 2022, from https://www.researchgate.net/publication/354400113_A_Random_Forest_Students%27_Performance_Prediction_RFSPP_Model_Based_on_Students%27_Demographic_Features

[10] Berens, J., Schneider, K., Görtz, S., Oster, S., & Burghoff, J. (2018, October 31). Early detection of students at risk – predicting student dropouts using administrative student data and machine learning methods. SSRN. Retrieved July 29, 2022, from https://deliverypdf.ssrn.com/delivery.php?ID=265092002101088123006087103102121111000052061006004009102015047004032118003009068023086071003023080075099079070003113000118086024100121020001111065126002069023024114067102117112021&EXT=pdf&INDEX=TRUE

[11] Chen, W. (2021, February). Learning analytics for inclusive higher education. ResearchGate. Retrieved July 29, 2022, from https://www.researchgate.net/profile/Weiqin-Chen-2/publication/351870067_Learning_Analytics_for_Inclusive_Higher_Education/links/60ae161aa6fdcc647edc387e/Learning-Analytics-for-Inclusive-Higher-Education.pdf

[12] Chicco, D., Warrens, M. J., & Jurman, G. (2021, July 5). The coefficient of determination R-squared is more informative than smape, Mae, MAPE, MSE and RMSE in regression analysis evaluation. PeerJ. Computer science. Retrieved July 29, 2022, from https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8279135/

[13] Kent, M., Ellis, K., & Giles, M. (2018, September 27). Students with disabilities and elearning in Australia: Experiences of accessibility and disclosure at Curtin University - TechTrends. SpringerLink. Retrieved July 29, 2022, from https://link.springer.com/article/10.1007/s11528-018-0337-y

[14] Kraayenoord, C. E. van, & Chapman, J. (2016, March). Learning disabilities in Australia and New Zealand . ResearchGate. Retrieved July 29, 2022, from https://www.researchgate.net/publication/303993211_Learning_Disabilities_in_Australia_and_New_Zealand

[15] Kuzilek J., Hlosta M., Zdrahal Z. Open University Learning Analytics dataset Sci. Data 4:170171 doi: 10.1038/sdata.2017.171 (2017)

[16] Lambert, D. C., & Dryer, R. (2017, December 26). Quality of life of higher education students with learning disability studying online. Taylor & Francis. Retrieved July 29, 2022, from https://www.tandfonline.com/doi/full/10.1080/1034912X.2017.1410876

[17] Pandey, A., & Jain, A. (2017, November). Comparative Analysis of KNN Algorithm using Various Normalization Techniques. Modern Education and Computer Science. Retrieved July 29, 2022, from https://j.mecs-press.net/ijcnis/ijcnis-v9-n11/IJCNIS-V9-N11-4.pdf

[18] Randhawa, K., Loo, C. K., Seera, M., Lim, C. P., & Nandi, A. K. (2018, February 15). Credit card fraud detection using ADABOOST and majority voting. IEEE Xplore. Retrieved July 29, 2022, from https://ieeexplore.ieee.org/abstract/document/8292883

[19] Riazy, S., & Simbeck, K. (1970, January 1). Predictive algorithms in learning analytics and their fairness. Startseite - Digitale Bibliothek - Gesellschaft für Informatik e.V. Retrieved July 29, 2022, from https://dl.gi.de/handle/20.500.12116/24401

[20] Rizvi, S., Rienties, B., & Khoja, S. A. (2019, April 5). The role of demographics in online learning; a decision tree based approach. Computers & Education. Retrieved July 29, 2022, from https://www.sciencedirect.com/science/article/pii/S0360131519300818?casa_token=TDXrdtG-uv4AAAAA%3Aw6T31BV2gfO0P2ScJ3AHyz7ahrkC_PXf4jBbSyHp3xkGx6nJNWkRH6EyFhw_1013VgwhrFTdFEFi

[21] Sherry , S. L. (2017, August 29). Computer-Assisted Mathematics Instruction for Students With Specific Learning Disability: A Review of the Literature. Journal of Special Education Technology. Retrieved July 29, 2022, from https://journals.sagepub.com/doi/10.1177/016264341772588l

[22] Yağcı, M. (2022, March 3). Educational Data Mining: Prediction of Students' academic performance using machine learning algorithms - smart learning environments. SpringerOpen. Retrieved July 29, 2022, from https://slejournal.springeropen.com/articles/10.1186/s40561-022-00192-z

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